

Rover correction-burst timing and telemetry transmit plan

race-tracker · RTK base and rover captures of 7-8 October 2026 · working notes for the radio schedule discussion

Purpose

Each yacht carries two 900 MHz radios: the ArduSimple correction radio (receiving RTK corrections from the base once a second) and the telemetry radio (sending position fixes back). A transmitting telemetry radio a metre or two from the correction receiver can overload it, and with 30-50 yachts the telemetry transmissions also collide with each other. This note records what the captures show about the correction burst, and the transmit rule that follows: **send telemetry only outside the correction burst**.

What we measured

Capture	What it is	Key result
Base, USB output (60 s)	PyGPSClient parsed log from the RTK base board	1 Hz, six messages per epoch (1005, 1074, 1084, 1094, 1124, 1230), MSM4 only, no gaps. About 496 bytes per epoch (calculated from the message layouts).
Rover, client log (59 s)	PyGPSClient log from the rover board, 2.5 m from the base	Corrections arrive perfectly: 354 messages, all used, 0 checksum failures. RTK float for the whole capture, about 9 fixes per second, weak signal levels (L1 about 32 dB-Hz).
Rover, app console (this note)	Boat app run with GPS_LOG=1 GPS_LOG_RTCM=1; each [rtcm] line is timestamped when the receiver reports the message	Burst timing within each second - see below. Six consecutive bursts.

Correction burst timing at the rover

Second	First message (ms after the second)	Last message (ms)	Width (ms)	Gaps between messages (ms)
01:24:53	144	228	84	36, 11, 7, 0, 30
01:24:54	144	270	126	56, 12, 5, 0, 53
01:24:55	117	246	129	69, 10, 48, 0, 2
01:24:56	118	247	129	68, 10, 47, 0, 4
01:24:57	142	259	117	46, 10, 14, 0, 47
01:24:58	114	246	132	52, 12, 6, 0, 62
Range	114-144 (mean 130)	228-270 (mean 249)	84-132 (mean 120)	largest 69

Message order is the same every second: 1074, 1084, 1094, 1124, 1230, 1005. Times are when the app logged each message (receiver to USB to Node), so a few milliseconds of latency and jitter are included.

What this tells us

- **The burst is short and steady.** It starts 114-144 ms after each second and is over by 270 ms, about 120 ms wide on average.
- **The correction radio is not on its slowest rate.** About 500 bytes arrived in roughly 0.1 s; a 10 kb/s link would need 400 ms or more for that.
- **Treat the whole cluster as one blocked period.** The radio delivers the messages in chunks, with gaps of up to 69 ms inside the burst, so transmitting in a gap could still hit the next chunk.

- **The timing is predictable.** The burst repeats once a second at a stable phase, so a rover can learn it from its own correction messages and predict the next one.

Proposed protected window

Using a 60 ms guard each side (UART and Node jitter, plus USB latency in these timestamps): block telemetry from **54 ms to 330 ms** after each second, a window of **276 ms**. That protects the burst and leaves about **724 ms (72%)** of every second for uplinks. The window should be learned continuously from the correction messages, not hard-coded, and should not be applied when no bursts are heard.

Telemetry load against the free time

Fleet	Fixes per second	Bytes per second	Share of free time
30 boats, all 2 Hz	60	1,260	15%
30 boats, 10 at 5 Hz (near the finish), 20 at 2 Hz	90	1,890	23%
30 boats, all 5 Hz	150	3,150	38%
50 boats, all 2 Hz	100	2,100	25%
50 boats, 15 at 5 Hz, 35 at 2 Hz	145	3,045	37%
50 boats, all 5 Hz	250	5,250	63%
30 boats, all at the full ~10 Hz	300	6,300	76%

Assumes 21 bytes per fix (batches of four, headers included) and an 11.5 kB/s usable channel (from the radio congestion test), of which 72% is free outside the burst. As far as the Digi user guide shows, the telemetry radio does no carrier sensing or retries in this configuration (unacknowledged broadcast), so random-access transmissions collide; shares above roughly 20% need staggered or assigned slots.

Plan

- **Burst tracker** in the boat app: group the receiver's correction messages into one burst per second, learn the start, end and phase, and predict the next burst.
- **Transmit gate:** hold a frame if sending it would overlap the predicted burst plus guard; open the gate when no bursts have been heard for a few seconds.
- **Staggered release:** release held frames at a per-rover offset in the free window (from the boat id), so rovers do not all transmit the instant the window ends. Assigned slots (the consultant's schedule) would remove the remaining collisions.
- **Flag and test:** behind a setting such as TX_AVOID_RTCM_BURST; first in the simulator with a replayed burst and 30 virtual rovers, then on a real rover comparing correction failures with the flag on and off while the congestion test runs.

Caveats

- Only six bursts were timed; a longer capture (a few minutes, with the rover at distance) should confirm the range and show any drift.
- Timestamps include USB and Node latency. The [gps] lines in the appendix are only those printed between correction lines (the console overwrites them in place), so they say nothing about the fix rate.
- The rover was 2.5 m from the base. Burst length at range, and with retries on a weak link, may differ.

Appendix: rover console log (verbatim)

Boat app output with GPS_LOG=1 GPS_LOG_RTCM=1, rover over USB, RADIO_ENABLED=0. Also saved as docs/rover-rtcm-burst-timing-2026-10-08.log.

```
[gps] 01:24:52.900 36.954617, -121.974636 fixType=3 diffSoln=true carrSoln=1 numSV=31 hAcc=0.01m
2026-10-08T01:24:53.144Z [rtcm] type=1074 station=0 used=used
2026-10-08T01:24:53.180Z [rtcm] type=1084 station=0 used=used
2026-10-08T01:24:53.191Z [rtcm] type=1094 station=0 used=used
2026-10-08T01:24:53.198Z [rtcm] type=1124 station=0 used=used
2026-10-08T01:24:53.198Z [rtcm] type=1230 station=0 used=used
[gps] 01:24:53.100 36.954617, -121.974636 fixType=3 diffSoln=true carrSoln=1 numSV=31 hAcc=0.01m
2026-10-08T01:24:53.228Z [rtcm] type=1005 station=0 used=used
[gps] 01:24:53.900 36.954617, -121.974636 fixType=3 diffSoln=true carrSoln=1 numSV=31 hAcc=0.01m
2026-10-08T01:24:54.144Z [rtcm] type=1074 station=0 used=used
2026-10-08T01:24:54.200Z [rtcm] type=1084 station=0 used=used
2026-10-08T01:24:54.212Z [rtcm] type=1094 station=0 used=used
[gps] 01:24:54.100 36.954617, -121.974636 fixType=3 diffSoln=true carrSoln=1 numSV=31 hAcc=0.01m
2026-10-08T01:24:54.217Z [rtcm] type=1124 station=0 used=used
2026-10-08T01:24:54.217Z [rtcm] type=1230 station=0 used=used
2026-10-08T01:24:54.270Z [rtcm] type=1005 station=0 used=used
[gps] 01:24:54.800 36.954617, -121.974636 fixType=3 diffSoln=true carrSoln=1 numSV=31 hAcc=0.01m
2026-10-08T01:24:55.117Z [rtcm] type=1074 station=0 used=used
[gps] 01:24:55.000 36.954617, -121.974636 fixType=3 diffSoln=true carrSoln=1 numSV=31 hAcc=0.01m
2026-10-08T01:24:55.186Z [rtcm] type=1084 station=0 used=used
2026-10-08T01:24:55.196Z [rtcm] type=1094 station=0 used=used
2026-10-08T01:24:55.244Z [rtcm] type=1124 station=0 used=used
2026-10-08T01:24:55.244Z [rtcm] type=1230 station=0 used=used
2026-10-08T01:24:55.246Z [rtcm] type=1005 station=0 used=used
[gps] 01:24:55.800 36.954617, -121.974636 fixType=3 diffSoln=true carrSoln=1 numSV=31 hAcc=0.01m
2026-10-08T01:24:56.118Z [rtcm] type=1074 station=0 used=used
[gps] 01:24:56.000 36.954617, -121.974636 fixType=3 diffSoln=true carrSoln=1 numSV=31 hAcc=0.01m
2026-10-08T01:24:56.186Z [rtcm] type=1084 station=0 used=used
2026-10-08T01:24:56.196Z [rtcm] type=1094 station=0 used=used
2026-10-08T01:24:56.243Z [rtcm] type=1124 station=0 used=used
2026-10-08T01:24:56.243Z [rtcm] type=1230 station=0 used=used
2026-10-08T01:24:56.247Z [rtcm] type=1005 station=0 used=used
[gps] 01:24:56.900 36.954617, -121.974636 fixType=3 diffSoln=true carrSoln=1 numSV=31 hAcc=0.01m
2026-10-08T01:24:57.142Z [rtcm] type=1074 station=0 used=used
2026-10-08T01:24:57.188Z [rtcm] type=1084 station=0 used=used
2026-10-08T01:24:57.198Z [rtcm] type=1094 station=0 used=used
2026-10-08T01:24:57.212Z [rtcm] type=1124 station=0 used=used
2026-10-08T01:24:57.212Z [rtcm] type=1230 station=0 used=used
[gps] 01:24:57.100 36.954617, -121.974636 fixType=3 diffSoln=true carrSoln=1 numSV=31 hAcc=0.01m
2026-10-08T01:24:57.259Z [rtcm] type=1005 station=0 used=used
[gps] 01:24:57.900 36.954617, -121.974636 fixType=3 diffSoln=true carrSoln=1 numSV=31 hAcc=0.01m
2026-10-08T01:24:58.114Z [rtcm] type=1074 station=0 used=used
[gps] 01:24:58.000 36.954617, -121.974636 fixType=3 diffSoln=true carrSoln=1 numSV=31 hAcc=0.01m
2026-10-08T01:24:58.166Z [rtcm] type=1084 station=0 used=used
2026-10-08T01:24:58.178Z [rtcm] type=1094 station=0 used=used
2026-10-08T01:24:58.184Z [rtcm] type=1124 station=0 used=used
2026-10-08T01:24:58.184Z [rtcm] type=1230 station=0 used=used
2026-10-08T01:24:58.246Z [rtcm] type=1005 station=0 used=used
```